

3 Shallow neural networks

3.1 If the activation σ is linear / the identity link,
the whole network collapses to an affine function.

3.3

candidate joints c_i where $h_i = 0$

$$\theta_{10} + \theta_{11}x = 0 \Rightarrow -\theta_{10}/\theta_{11} = c_1$$

$$\theta_{20} + \theta_{21}x = 0 \Rightarrow -\theta_{20}/\theta_{21} = c_2$$

$$\theta_{30} + \theta_{31}x = 0 \Rightarrow -\theta_{30}/\theta_{31} = c_3$$

As no unit h_i ($i \in \{1, 2, 3\}$) has a corresponding $\phi_i = 0$,

and, from the picture, all c_i are different,

the set of all joints is: $\{-\theta_{10}/\theta_{11}, -\theta_{20}/\theta_{21}, -\theta_{30}/\theta_{31}\}$

The slopes are

$$[0, -\theta_{10}/\theta_{11}] : \text{only } h_3 \text{ active: } \phi_0 + \phi_3 h_3 = y = \phi_0 + \phi_3 \theta_{30} + \phi_3 \theta_{31}x$$

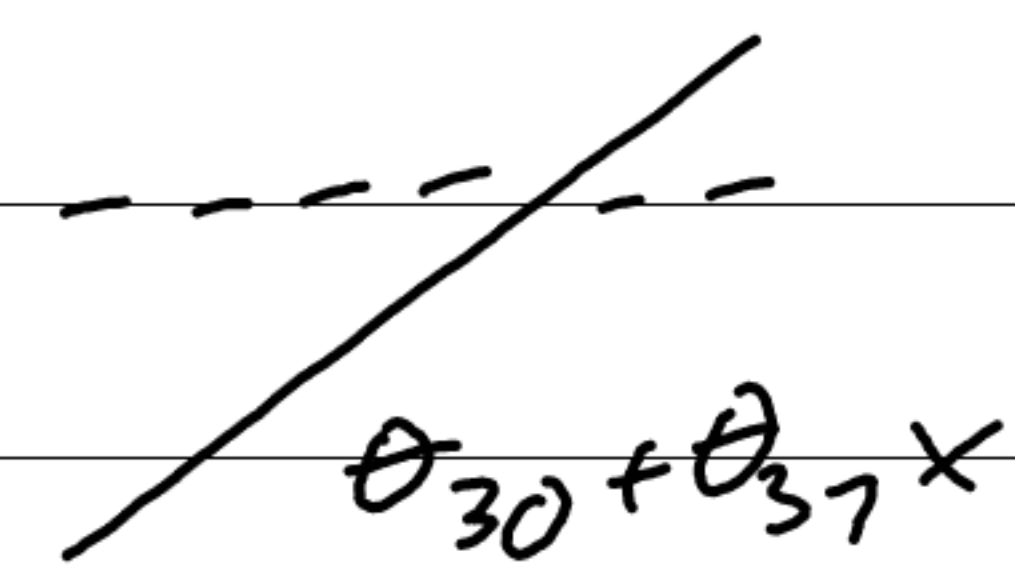
$$\text{slope: } \frac{dy}{dx} = \phi_3 \theta_{31}$$

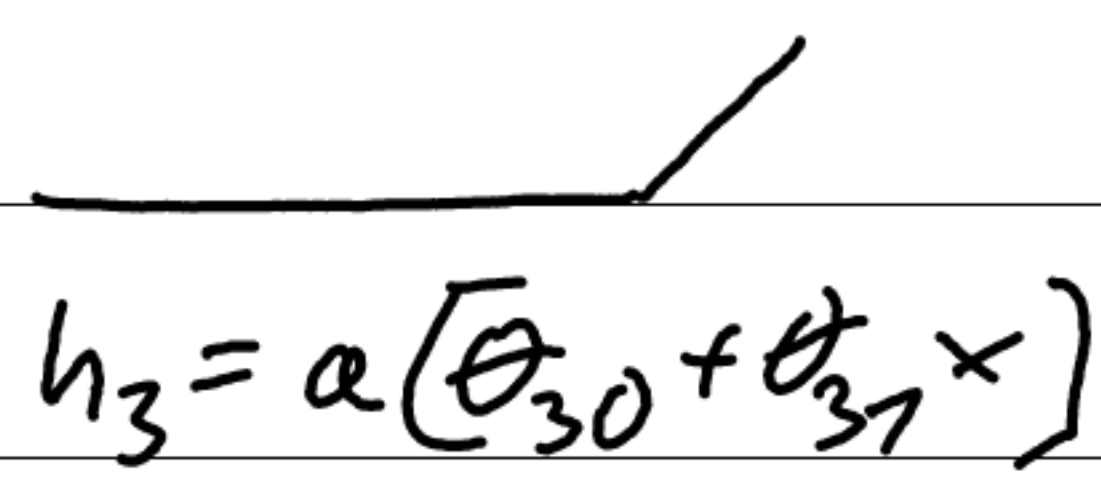
$$[-\theta_{10}/\theta_{11}, -\theta_{20}/\theta_{21}] : h_1, h_3 \text{ active: } y = \phi_0 + \phi_1 \theta_{10} + \phi_1 \theta_{11}x + \phi_3 \theta_{30} + \phi_3 \theta_{31}x$$

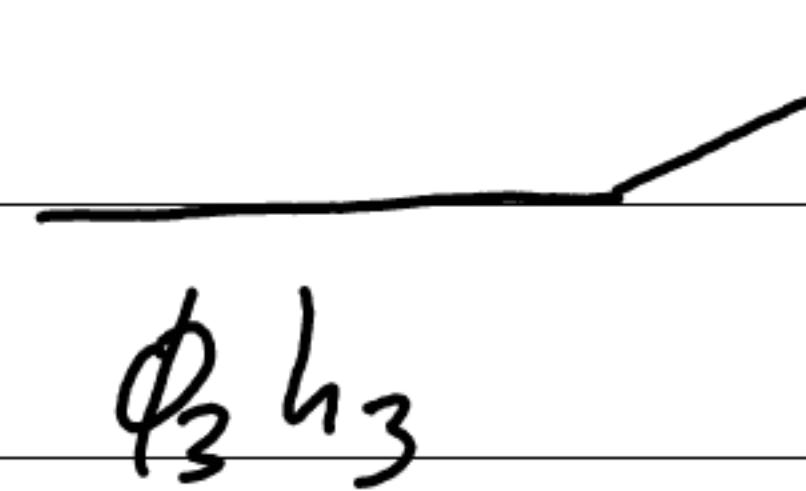
$$\text{slope: } \frac{dy}{dx} = \phi_1 \theta_{11} + \phi_3 \theta_{31}$$

$$[-\theta_{20}/\theta_{21}, -\theta_{30}/\theta_{31}] : h_1, h_2, h_3 \text{ active: } \frac{dy}{dx} = \phi_1 \theta_{11} + \phi_2 \theta_{21} + \phi_3 \theta_{31}$$

$$[-\theta_{30}/\theta_{31}, 2] : h_1, h_2 \text{ active: } \frac{dy}{dx} = \phi_1 \theta_{11} + \phi_2 \theta_{21}$$

3.4 c) 

f) 

i) 

j)



3.5 $\text{ReLU}(az) = a \text{ReLU}(z) \quad \forall a \in \mathbb{R}^+$

proof: $\text{ReLU}(az) = \max(0, az)$

case 1: $z \geq 0$: $\text{ReLU}(az) = az = a \text{ReLU}(z)$

case 2: $z < 0$: $\text{ReLU}(az) = 0 = a \cdot 0 = a \text{ReLU}(z) \quad \square$

3.6 If α is positive, nothing happens:

$$y = \phi_0 + \underbrace{\phi_1 h_1}_{\frac{1}{2} \phi_1 a[\alpha \theta_{10} + \alpha \theta_{11}x]} + \phi_2 h_2 + \phi_3 h_3$$

$$\frac{1}{2} \phi_1 a[\alpha \theta_{10} + \alpha \theta_{11}x] = \frac{1}{2} \phi_1 a[\alpha (\theta_{10} + \theta_{11}x)]$$

$$\alpha > 0 \Rightarrow \frac{1}{2} \phi_1 a[\theta_{10} + \theta_{11}x]$$

If $\alpha < 0$: we cannot use 3.5, as we would swap the clipped / unclipped regions. Additionally, multiplying h_1 by $\frac{1}{2}$ would result in negative slope (however, leaving the slope of $\phi_1 \theta_{11}$ unchanged).

3.7 As we've shown in 3.6, ReLU activations

break scale invariances:

$$\frac{1}{\alpha} \text{ReLU}(\alpha z) = \text{ReLU}(z).$$

Thus, we can find e.g., $\phi_1, \theta_{10}, \theta_{11}$

$$\text{so that } \phi_1' \alpha [\theta_{10}' + \theta_{11}' x] = \phi_1 \alpha [\theta_{10} + \theta_{11} x]$$

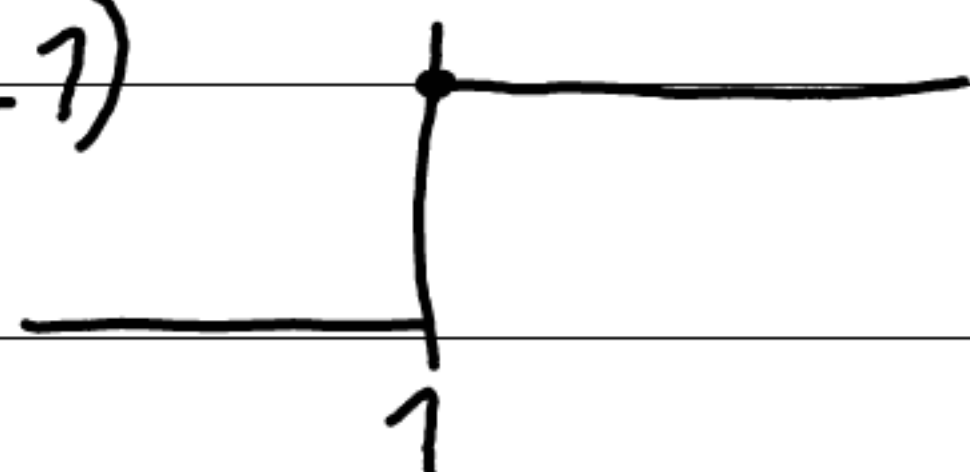
with $\phi_1' \neq \phi_1, \dots$, meaning different solutions

(denoting $\{\phi_1, \theta_{10}, \theta_{11}\}$ as a unique best set of param.)

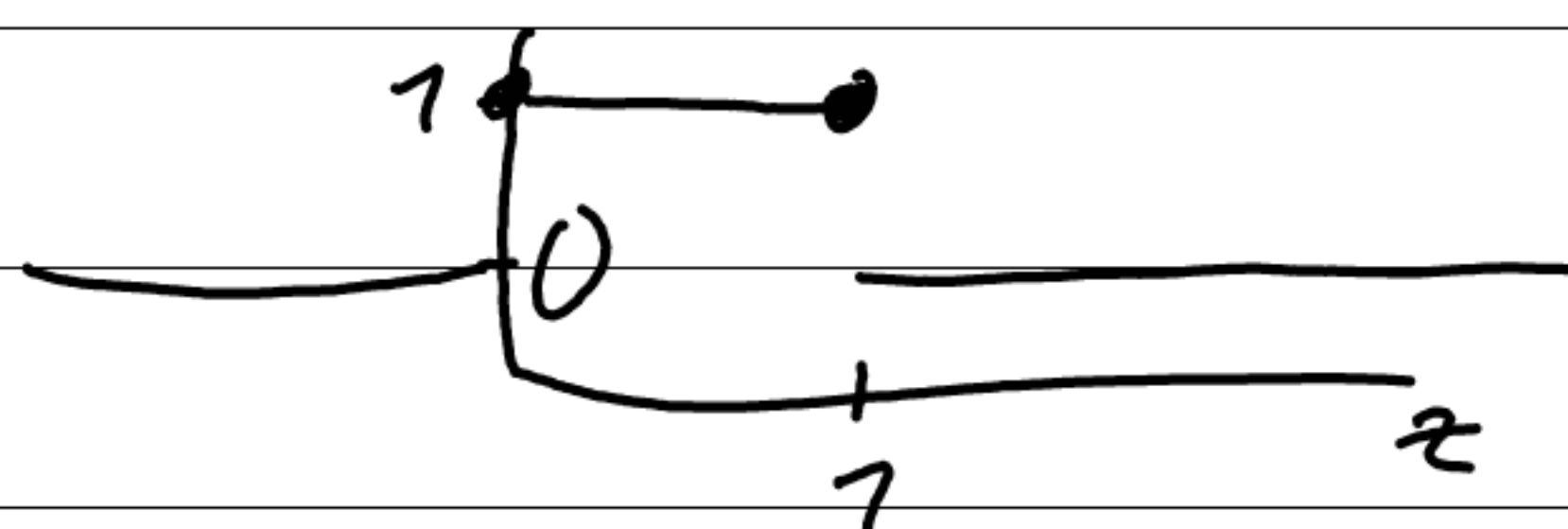
"L2 is convex in predictions, but not necessarily convex in parameters once you restrict predictions to come from a neural network".

3.8 $\text{heaviside}[z] = \begin{cases} 0 & z < 0 \\ 1 & z \geq 0 \end{cases}$

$\mathbb{I}(x \geq 1)$



$$\text{rect}[z] = \begin{cases} 0 & z < 0 \\ 1 & 0 \leq z \leq 1 \\ 0 & z > 1 \end{cases}$$



$$\text{tanh}[z] = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

see [prince/exercise/0308.ipynb](#)

family of Archias: tanh: wiggy s-convolutions (up to 3 bumps)

heaviside: step functions with up to 4 steps $\xrightarrow{h_1} \xrightarrow{h_2} \xrightarrow{h_3}$

rect: step functions with up to 6 steps (3 bumps)!

3.9 slope 3rd = slope 1st + slope 4th

3rd region: h_1, h_2, h_3 active

1st region: h_3 active

2nd region: h_1, h_2 active

$$\begin{aligned}\frac{\partial}{\partial x}(\phi_0 + \phi_1 h_1 + \phi_2 h_2 + \phi_3 h_3) &= \frac{\partial}{\partial x} \phi_1 h_1 + \frac{\partial}{\partial x} \phi_2 h_2 + \frac{\partial}{\partial x} \phi_3 h_3 \\ &= \frac{\partial}{\partial x}(\phi_3 h_3) + \frac{\partial}{\partial x}(\phi_1 h_1 + \phi_2 h_2) \quad \square\end{aligned}$$

3.10 If a hidden unit never activates, e.g. for ReLU, it stays under 0 from $[0, 2]$.

ALSO if 2 units switch at the same x , and their slopes cancel (see 3.9), no additional region is created!

3.11 18 parameters (fig 3.6)

3.12 $9 + 4 = 13$ params. (fig 3.7)

3.13 see 1N

$$3.14 \quad y_1 := \phi_{10} + \phi_{11} h_1 + \phi_{12} h_2 + \phi_{13} h_3$$

$$y_2 := \phi_{20} + \phi_{21} h_1 + \phi_{22} h_2 + \phi_{23} h_3$$

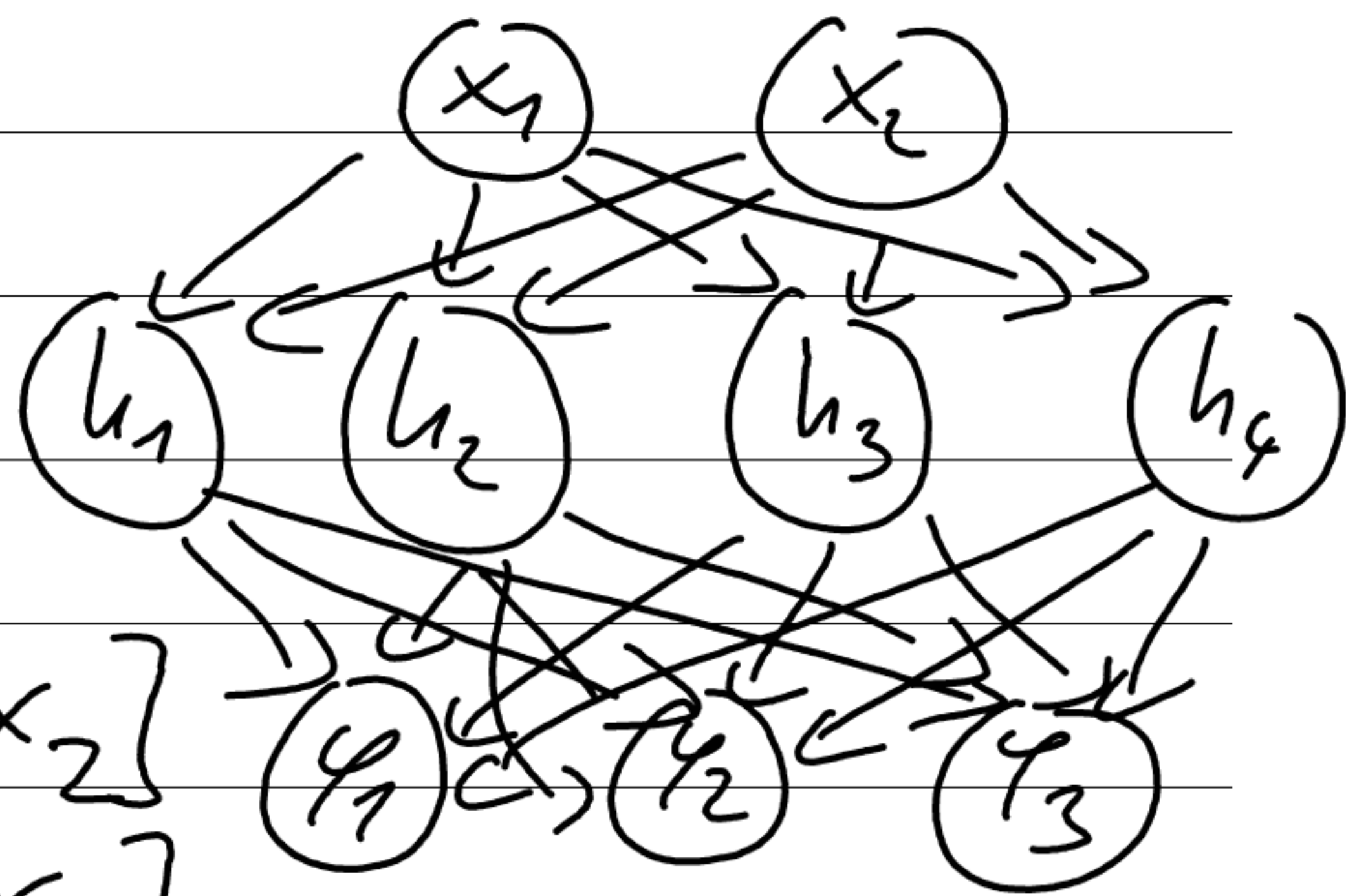
$$h_1 := a[\theta_{10} + \theta_{11} x_1 + \theta_{12} x_2 + \theta_{13} x_3]$$

$$h_2 := a[\theta_{20} + \theta_{21} x_1 + \theta_{22} x_2 + \theta_{23} x_3]$$

$$h_3 := a[\theta_{30} + \theta_{31} x_1 + \theta_{32} x_2 + \theta_{33} x_3]$$

$$3.15 \quad \binom{3}{0} + \binom{3}{1} + \binom{3}{2} + \binom{3}{3} = 8$$

$$3.16 \quad \begin{aligned} y_1 &= \phi_{10} + \sum_{d=1}^4 \phi_{1d} h_d \\ y_2 &= \phi_{20} + \sum_{d=1}^4 \phi_{2d} h_d \\ y_3 &= \phi_{30} + \sum_{d=1}^4 \phi_{3d} h_d \end{aligned}$$



$$h_1 = a[\theta_{10} + \theta_{11}x_1 + \theta_{12}x_2]$$

$$h_2 = a[\theta_{20} + \theta_{21}x_1 + \theta_{22}x_2]$$

$$h_3 = a[\theta_{30} + \theta_{31}x_1 + \theta_{32}x_2]$$

$$h_4 = a[\theta_{40} + \theta_{41}x_1 + \theta_{42}x_2]$$

3.17 D_i inputs

D hidden units

D_o outputs

$$|\text{parameters}| = D_i D + D_o D + D + D_o$$

$$= D_i D + 1 \times D + D_o D + D_o \times 1$$

$$= (D_i + 1) D + D_o (D + 1)$$

$$3.18 \quad N = \binom{3}{0} + \binom{3}{1} + \binom{3}{2} = 7$$

$$N = \binom{5}{0} + \binom{5}{1} + \binom{5}{2} = 16$$